Prompt-Based Collaborative Agent Framework (PCAF) for Enhanced LLM Mathematical Reasoning

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Abstract

Summarize the goal (improving final-answer accuracy on MathArena), the method (single LLM, 3 roles: Solver, Verifier, Planner), and the key result (performance vs. Zero-Shot CoT baseline).

The inherent unreliability of Large Language Models (LLMs) in complex, multi-step calculations necessitates architectural interventions to achieve trustworthy accuracy in mathematical reasoning. This paper introduces the Prompt-Based Collaborative Agent Framework (PCAF), a novel, architecture-driven approach aimed at significantly improving final-answer accuracy on challenging benchmarks such as MathArena (including AIME 2025 and HMMT Deb 2025 subsets). The method utilizes a single LLM instance (DeepSeek-V3-03-24) and employs sequential, role-specific prompting to simulate a sophisticated, three-agent collaborative system. The framework operates in a continuous, closed-loop refinement cycle defined by three specialized agents: a Solver, which generates the solution under a strict Dual-Method Verification and Code-First Mandate; a Verifier, which acts as a Skeptical Math Auditor to generate a structured JSON critique; and a Planner, which translates the critique into an explicit, non-negotiable MANDATORY ACTION instruction for the next attempt. This architecture transforms the LLM's linear reasoning into a persistent self-correction mechanism. Through systematic evaluation, we establish a robust baseline using Zero-Shot Chain-of-Thought (CoT) and demonstrate that the PCAF's structured intervention yields a significant uplift in final accuracy, confirming the efficacy of deterministic, architecture-driven reflection over traditional self-correction techniques.
Results are not included yet

1 Introduction

What to write:

- The Problem: LLMs struggle with precise calculation and are "stubborn" in self-correction.
- The Approach: Instead of fine-tuning, you use architecture-driven reasoning. Introduce the PCAF concept (sequential, role-specific prompting).
- Contributions:
- The novel algorithmic structure (Solver/Verifier/Planner loop).

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

- The use of a Python Sandbox to enforce "grounding" in code.
- Targeted improvement on the MathArena benchmark.

The pursuit of reliable mathematical reasoning in Large Language Models (LLMs) is hindered by two persistent challenges: computational imprecision and stubbornness in self-correction. While LLMs excel at generating coherent solution narratives (Chain-of-Thought), they frequently fail at the final, precise steps—arithmetic, algebra, and complex counting—which leads to high rates of final-answer error on benchmarks like MathArena. Furthermore, models typically struggle to effectively diagnose and pivot from their initial logical errors, making standard self-correction techniques unreliable.
To overcome these limitations, this project implements the Prompt-Based Collaborative Agent Framework (PCAF). Instead of relying on expensive model fine-tuning, PCAF utilizes an architecture-driven reasoning approach. It simulates a multi-agent system within a single LLM instance through sequential, role-specific prompting. The framework dynamically cycles the LLM through three specialized roles—Solver, Verifier, and Planner—in an iterative, closed-loop refinement process. The core objective is to demonstrate that this structured collaborative architecture significantly increases the final-answer accuracy and trustworthiness compared to the model's standard Zero-Shot baseline. The key contributions of this work are:
Novel Algorithmic Structure: The PCAF loop itself, which formalizes the iterative roles of Generation (Solver), Critique (Verifier), and Coordination (Planner), forcing systematic logical scrutiny.
Enforced Grounding: Mandatory integration of a persistent Python sandbox environment ("Code" tool) to enforce Dual-Method Verification (Analytical vs. Naive Brute Force), ensuring mathematical conclusions are derived from verifiable, executed code.
Targeted Improvement: The application and evaluation of PCAF on the challenging final-answer sub-sections of the MathArena benchmark (AIME 2025, HMMT Feb 2025 Final-Answer Competitions) to provide a clear, quantitative measure of accuracy uplift.

2 Related Work

The Prompt-Based Collaborative Agent Framework (PCAF) is informed by two established and growing areas of research in LLMs: Reflective Reasoning and Verification and Multi-Agent Systems for Problem Solving.

2.1 Reflective Reasoning and Verification

Early methods in mathematical reasoning focused on generating a single, detailed Chain-of-Thought (CoT). However, recent work emphasizes enhancing reliability through self-critique. Approaches such as those presented by Wang et al. (2025) [3], who introduce a collaboration mechanism for long CoT reasoning, and Li et al. (2025) [1], who explores LLM agents for automated proof generation using knowledge graphs, highlight the value of structured introspection. These systems often succeed in generating multiple solution paths or utilizing formal methods to check internal consistency. However, a common limitation is the nature of the corrective feedback; the system may identify an error but lack a dedicated, high-level mechanism to effectively re-plan the solution strategy. The feedback loop often remains constrained to simple re-prompts or voting systems.

2.2 Multi-Agent Systems and Formal Verification

A parallel line of inquiry involves distributing tasks among distinct, specialized LLM instances to simulate collaborative environments. Works like Varambally et al. (2025) [2], which describe a system for recursively building formal proofs, employ specialized agents (a "formal prover" and a "critic") to ensure soundness. This multi-model approach is computationally powerful and minimizes agent bias but requires the coordination and hosting of multiple distinct models. Furthermore, when these systems interact with external tools, such as the formal prover used , the communication requires complex integration layers.

3 Methodology

3.1 Framework Overview

Describe the iterative loop ($N = 3$ max iterations). The PCAF is a deterministic, iterative, and closed-loop system designed to enforce self-correction and dual-verification. The process is initiated with the problem statement and runs for a maximum of $N = 4$ total attempts (one initial attempt plus three retries). The core cycle involves three sequential steps:
1. Attempt (Solver): The Solver generates a solution and its associated execution trace.
2. Verification (Verifier): The Verifier audits the complete trace against a structured checklist and returns a structured critique.
3. **Correction/Re-plan (Planner $\rightarrow$ Solver):** If the Verifier rejects the solution, the Planner translates the critique into an explicit, targeted instruction that overrides the Solver's default behavior, forcing a specific corrective action in the next attempt.The loop terminates successfully if the Verifier accepts the solution, or unsuccessfully if the maximum number of attempts is reached.

3.2 The Agents (Cite specific prompts from your code):

The framework's robustness derives from the dynamic specialization of the single LLM into three distinct agents via role-specific system prompts.

3.2.1 The Solver

The Solver is the generative component, constrained by a strict system prompt `get_solver_prompt` that enforces the Dual-Verification Protocol for grounding and reliability.

"Two Strictly Different Methods" Mandate: The Solver must use two conceptually distinct approaches to arrive at the same answer:

- **Analytical Method (Method 1):** Utilizes formal mathematics, algebra, or efficient libraries (e.g., `sympy`).
- **Naive Brute-Force (Method 2):** Requires a simple, unoptimized Python script to simulate or iterate through all possibilities.

"Code-First" Mandate: The Solver is required to execute the code for both methods and explicitly compare the outputs, ensuring the final answer is a value confirmed by an external tool. This process is crucial for preventing numerical hallucinations.

3.2.2 The Verifier

The Verifier's role is that of a Skeptical Math Auditor (`VERIFIER_ROLE`). It receives the Solver's full reasoning and code outputs and audits the solution based on a specific **Audit Checklist** (e.g., consistency between text and code, presence of the dual-method check, and logic trap checks).

The Verifier is strictly constrained to output a structured JSON critique that adheres to a predefined schema:

```
{JSON Schema:  "valid":  bool, "error_type":  str, "critique":
str }
```

The `error_type` field serves as the primary, discrete signal for the Planner, classifying failures into three key categories:

`LOGIC_FLAW` A fundamental error in the mathematical approach or application of a principle.

`METHOD_CONFLICT` The outputs of the two mandated methods (Analytical vs. Naive) did not match, or the comparison step was skipped.

`VALUE_MISMATCH` The logic or code was mostly correct, but the final textual answer did not match the computed output (e.g., failed to apply a final transformation like modulo or rounding).

3.2.3 The Planner

The Planner is not a separate LLM call but rather a deterministic, heuristic-driven function (`construct_planner_feedback`) that translates the Verifier's JSON rejection into a specific, high-leverage instruction for the subsequent Solver attempt.

The Planner's logic ensures that corrections are targeted and strategy-shifting:

`Targeted Reset for VALUE_MISMATCH` If the `VALUE_MISMATCH` flag is raised, the Planner generates an instruction to "Wipe memory of the 'known' answer and re-derive it from scratch using code." This counteracts the LLM's tendency to stubbornly repeat a flawed answer despite being shown the correct number in the trace.

`Strategy Shift for CONCEPTUAL_FLAW` For a conceptual failure in geometric problems, the Planner mandates a shift, e.g., "Switch to Coordinate Geometry. Place a vertex at (0,0) and use the Shoelace Formula."

This architecture ensures the correction loop is not a generic "try again," but a precise, planned intervention.

3.3 Tool Use

The execution of mathematical code is handled by a custom, stateful environment, the `PersistentSolverSandbox`. This tool is essential for providing verifiable grounding:
Persistent Environment: Unlike typical stateless code execution tools, the sandbox maintains its global state (`self.globals`) across consecutive code blocks within a single Solver attempt. This enables complex, multi-block workflows in which a function is defined in one block and then invoked or modified in subsequent blocks.

Comprehensive Imports: The sandbox is pre-loaded with essential mathematical and symbolic libraries (`math`, `sympy`, `numpy`) to support both analytical and naive solution methods.

Robust I/O Capture: The sandbox reliably captures all standard output (`stdout`) and handles common runtime issues (e.g., infinite loops via timeouts, `IndentationError`, and `SyntaxError`), returning any error messages to the Solver as an `OBSERVATION`.
This feedback loop iscritical, as it allows the Solver to debug its own code syntax and execution failures.

The Solver: Reference get$_{s}olver_{p}rompt.Explainthe"TwoStrictlyDifferentMethods"mandate(Analyticalvs.NaiveBr$ $Force)andthe"Code - FirstMandate".$

The Verifier: Reference VERIFIER$_{R}OLEandVERIFIER_{J}SON_{S}CHEMA.Explainhowitactsasa"SkepticalMathAud$

The Planner: Reference construct$_{p}lanner_{f}eedback.Explainthedeterministicroutinglogic.HighlighthowittranslatesaJS$ $Wipememory, re - calculate_{\text{J}}).$

Reference PersistentSolverSandbox. Explain how you use a persistent Python environment to maintain state variables between code blocks, handling imports (sympy, numpy) and capturing stdout.

4 Experimental setup

The evaluation of the Prompt-Based Collaborative Agent Framework (PCAF) was conducted using problems from the MathArena benchmark, specifically focusing on advanced "Final-Answer" subsets like AIME 2025 and HMMT Feb 2025 to ensure high problem complexity. The core reasoning engine was the DeepSeek-V3-03-24 Large Language Model, which was accessed via an API endpoint hosted on the Azure AI Foundry environment. To establish the gain from the PCAF architecture, the Zero-Shot Chain-of-Thought (CoT) performance (Solver agent only, no feedback loop) was measured as the baseline. The primary evaluation metric was Exact Match Accuracy, computed using the custom function check_correctness, which ensures semantic equivalence by handling numerical differences and symbolic simplifications (e.g., $1/2$ versus 0.5), thus accurately reflecting mathematical correctness.

Content: Dataset: MathArena (specifically the subsets loaded: HMMT, AIME, etc.). Mention you focused on the "Final Answer" subsets.

Model: DeepSeek-V3-03-24 hosted on Azure AI Foundry.Baselines: Zero-Shot Chain-of-Thought (Solver only, no feedback loop).

Evaluation Metric: Exact match accuracy using your check$_{c}orrectnessfunction(whichhandlessemanticequivalence, e.g., 1$

5 Results

What to report:

Quantitative Comparison: Create a table comparing Baseline Accuracy vs. PCAF Accuracy.

PCAF Performance: You need to calculate the final accuracy of PCAF from the pcaf$_{r}esults_{d}f.Evenifthetotalaccuracyissimilar, highlighttheRecoveredProblems.$

Recovery Rate: Highlight specific problems where attempts$_{u}sed > 1andis_{c}orrect == True.$

Example from logs: Problem ID 3 took 3 attempts. Problem ID 4 took 4 attempts. These are "wins" where the baseline failed, but the agents fixed it.

Average Turns: Report the average number of attempts required to reach a correct solution.

6 Analysis & Discussion

Successful Correction Traces: Walk through one example (e.g., Problem ID 4 from your logs). - Show: Solver failed (Attempt 1) $\rightarrow$ Verifier flagged METHOD$_{C}ONFLICT \rightarrow$ Planner forced a re-calculation $\rightarrow$ Solver fixed it.

Failure Analysis: Why did it fail even with retries? (Refer to the Rubric in your notebook).

Value Mismatch loops: (See Problem ID 30 in logs). The model kept calculating one thing but writing another.Geometric Hallucination: (See Problem ID 25 in logs). The Verifier flagged LOGIC$_{F}LAWregardinggeometricproperties, buttheSolvercouldn'tcorrectthecoordinategeometry.$

Impact of the Sandbox: Discuss how PersistentSolverSandbox prevented "hallucinated calculations" by forcing the model to actually run the code.

7 Conclusion

Summarize findings. Did structured collaboration improve reasoning? (Yes, it rescued several problems). Acknowledge limitations (e.g., sometimes the Verifier is too strict or the Planner gets stuck in a loop).

8 Abstract (THis is not how to do the abstract)

System Prompts: Copy the full text of SOLVER$_{R}OLE, VERIFIER_{R}OLE, andtheconstruct_{p}lanner_{f}eedbacklogichere.T$

Full Trace Example: Copy the JSON history of one full successful correction loop (e.g., Problem ID 3 or 4) to show the interaction.

8.1 Style

Papers to be submitted to NeurIPS 2025 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. Additional pages *containing references, checklist, and the optional technical appendices* do not count as content pages. Papers that exceed the page limit will not be reviewed, or in any other way considered for presentation at the conference.

The margins in 2025 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for rejection.

8.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

`https://neurips.cc`

The file `neurips_2025.pdf` contains these instructions and illustrates the various formatting requirements your NeurIPS paper must satisfy.

The only supported style file for NeurIPS 2025 is `neurips_2025.sty`, rewritten for $\LaTeX\,2_{\varepsilon}$. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported!**

The LaTeX style file contains three optional arguments: `final`, which creates a camera-ready copy, `preprint`, which creates a preprint for submission to, e.g., arXiv, and `nonatbib`, which will not load the `natbib` package for you in case of package clash.

Preprint option If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text "Preprint. Work in progress." in the footer. This version may be distributed as you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

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The file `neurips_2025.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 9, 10, and 11 below.

9 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 11 regarding figures, tables, acknowledgments, and references.

10 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

10.1 Headings: second level

Second-level headings should be in 10-point type.

10.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

11 Citations, figures, tables, references

These instructions apply to everyone.

11.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

> Hasselmo, et al. (1995) investigated...

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2025` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2025}
```

As submission is double blind, refer to your own published work in the third person. That is, use "In the previous work of Jones et al. [4]," not "In our previous work [4]." If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form "A. Anonymous" and include a copy of the anonymized paper in the supplementary material.

11.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[2] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[3]

11.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction. The figure number and caption always appear after the figure. Place one line space before the figure

[2] Sample of the first footnote.
[3] As in this example.

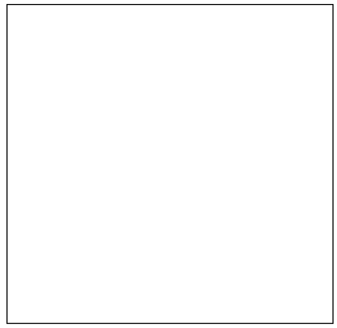

Figure 1: Sample figure caption.

Table 1: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

caption and one line space after the figure. The figure caption should be lower case (except for first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

11.4 Tables

All tables must be centered, neat, clean and legible. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 1.

11.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

11.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

12 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

12.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

Acknowledgments and Disclosure of Funding

Use unnumbered first level headings for the acknowledgments. All acknowledgments go at the end of the paper before the list of references. Moreover, you are required to declare funding (financial activities supporting the submitted work) and competing interests (related financial activities outside the submitted work). More information about this disclosure can be found at: `https://neurips.cc/Conferences/2025/PaperInformation/FundingDisclosure`.

Do **not** include this section in the anonymized submission, only in the final paper. You can use the `ack` environment provided in the style file to automatically hide this section in the anonymized submission.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Technical Appendices and Supplementary Material

Technical appendices with additional results, figures, graphs and proofs may be submitted with the paper submission before the full submission deadline (see above), or as a separate PDF in the ZIP file below before the supplementary material deadline. There is no page limit for the technical appendices.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes] , [No] , or [NA] .
- [NA] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for NA).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While "[Yes] " is generally preferable to "[No] ", it is perfectly acceptable to answer "[No] " provided a proper justification is given (e.g., "error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering "[No] " or "[NA] " is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

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- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer NA means that the paper does not include theoretical results.
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   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
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References

[1] Vincent Li, Yule Fu, Tim Knappe, Kevin Han, and Kevin Zhu. Automating mathematical proof generation using large language model agents and knowledge graphs, 2025. arXiv preprint.

[2] Sumanth Varambally, Thomas Voice, Yanchao Sun, Zhifeng Chen, Rose Yu, and Ke Ye. Hilbert: Recursively building formal proofs with informal reasoning, 2025. arXiv preprint.

[3] Ruida Wang, Rui Pan, Yuxin Li, Jipeng Zhang, Yizhen Jia, Shizhe Diao, Renjie Pi, Junjie Hu, and Tong Zhang. Ma-lot: Model-collaboration lean-based long chain-of-thought reasoning enhances formal theorem proving, 2025. arXiv preprint.